

1 Week 22: 31. January 2025

1.1 Part from ostium point estimation results

The estimations for five test set scans are investigated further in Figure 1. The first two represent the best cases, the next two average cases, and the last is the worst case. The top row shows the aligned source (light grey) with the target heatmap (blue), while the bottom row displays the source surface with the estimated sampled and best-guess aortic ostium points along with the LCA centerline. The general area where the LCA should originate is marked with a circle in the top row. In the best cases (two leftmost), the best-guess estimation is nearly spot on, with sampled estimates dispersed around the area. However, the best guess is further away in the average cases (middle and slightly right), nonetheless, few sampled estimates are closer to the LCA root. The increased distance between estimates and the LCA root is also attributed to an increased distance between the annotated LCA and aorta surface (seen in the slightly right case). In the worst case (rightmost), the best guess and most sampled estimates are far from an anatomically correct aortic ostium placement, though, a few samples are located in the desired general area, but still distant from the centerline root. Misalignment between the aligned source and target appears to be linked to poor estimates.

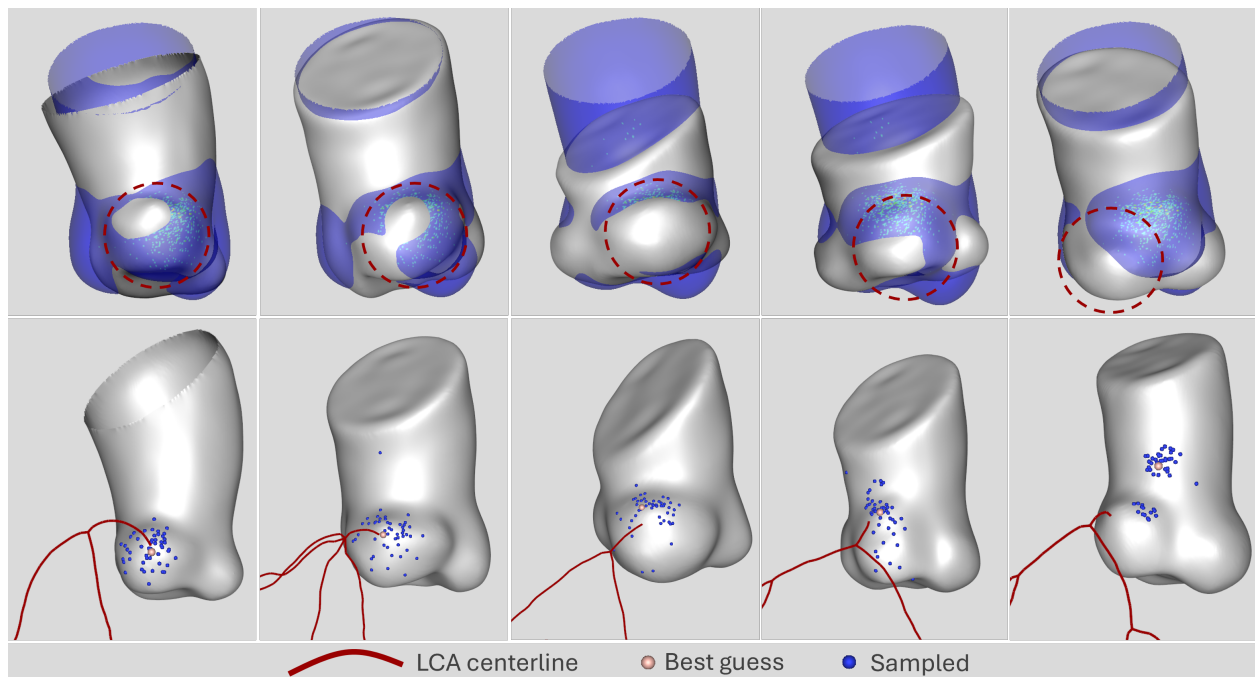


Figure 1: Illustration of aortic ostium point estimates for five scans (ImageCAS, LTR ids: 954.img, 939.img, 968.img, 936.img, 989.img). Top row: target aorta heatmap and aligned source, red circle marks the aortic ostium area. Bottom row: source aorta with best-guess (pink) and 50 sampled (blue) estimates obtained with prior method. Red curve is the ground truth LCA centerline.

It seems a bit strange that the aligned source aorta shape appears squeezed compared to the original one?

Investigating Heatmap Predictions

This appendix examines the causes of false positive bifurcation point detections (using the max-consensus model) through a series of examples on test set images. All examples show the ventricles, myocardium of the LV, and aorta segmentation along with the centerline for orientation purposes. Figures C.1-C.4 present cases where a predicted false positive bifurcation point may correspond to a tiny, non-annotated branch, suggesting that some false positives could be genuine bifurcations overlooked during annotation. However, the partial volume effect makes it challenging to determine whether these structures are truly tiny branches. Figure C.5 illustrates a case where a different structure passes closely by, possibly leading the model to mistakenly detect a bifurcation point. Figure C.6 shows an example of an incorrectly annotated LCA, causing a FP and FN. In reality, these correspond to a true bifurcation point and a non-bifurcation point, respectively, as predicted by the detector model.

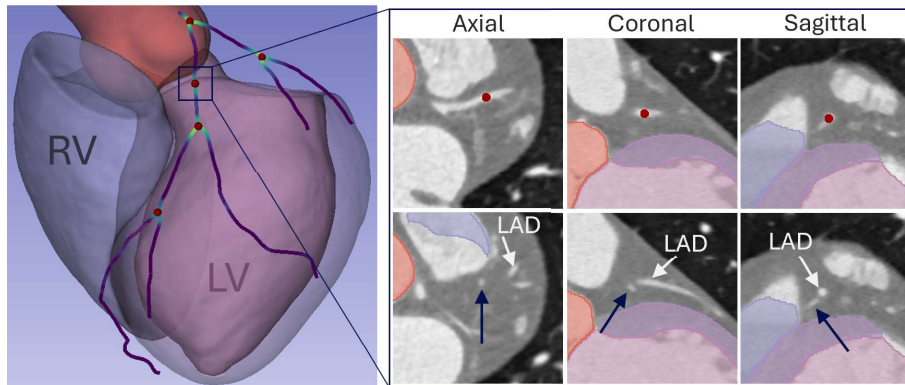


Figure C.1: Possibly tiny branch at FP prediction (ImageCAS, id: 942.img). Top: zoomed-in slices at the predicted point (red dot); bottom: adjacent slices. Blue arrows indicate a non-annotated branch.

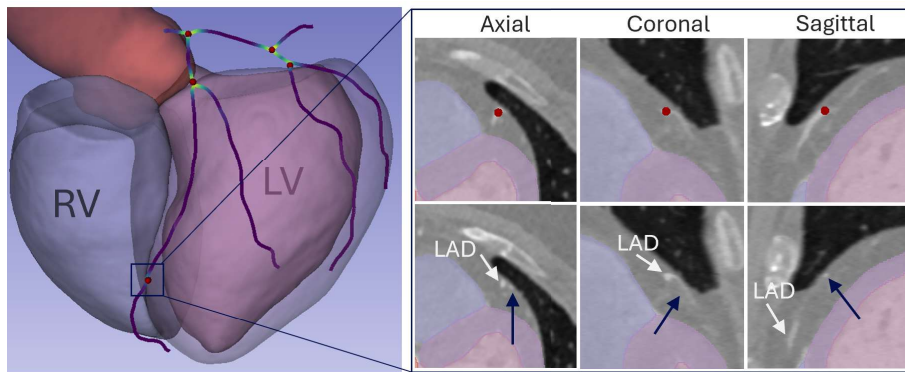


Figure C.2: Possibly tiny branch at FP prediction (ImageCAS, id: 996.img). Top: zoomed-in slices at the predicted point (red dot); bottom: adjacent slices. Blue arrows indicate a non-annotated branch.

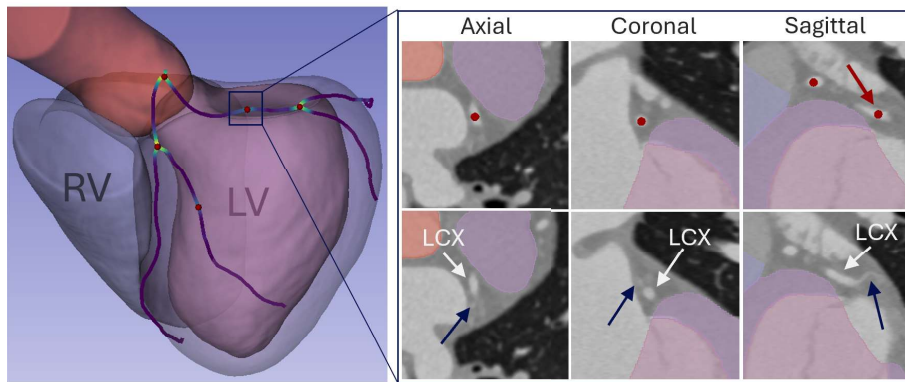


Figure C.3: Possibly tiny branch at FP prediction (ImageCAS, id: 938.img). Top: zoomed-in slices at the predicted point (red dot); bottom: adjacent slices. Blue arrows indicate a non-annotated branch.

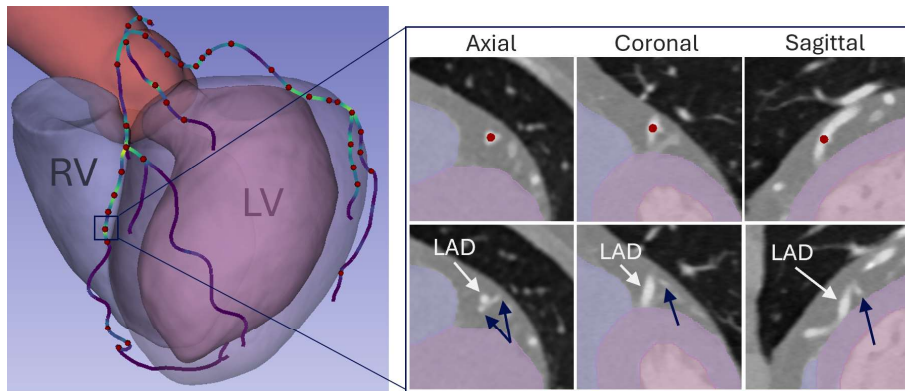


Figure C.4: Possibly tiny branch at FP prediction (ImageCAS, id: 974.img). Top: zoomed-in slices at the predicted point (red dot); bottom: adjacent slices. Blue arrows indicate a non-annotated branch.

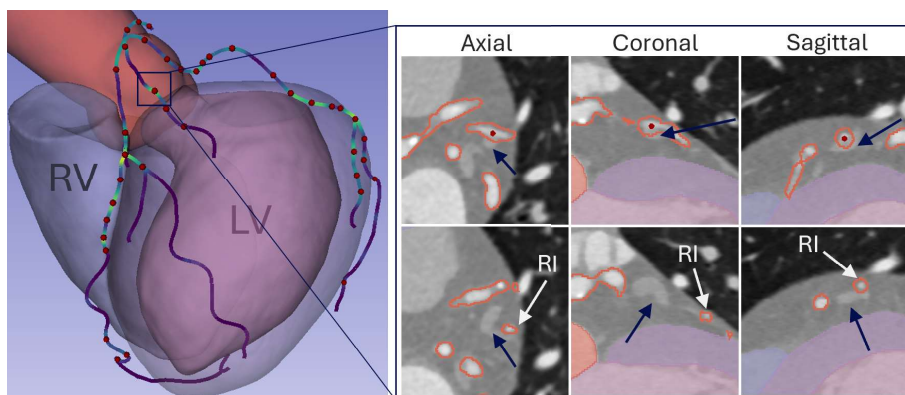


Figure C.5: Different structure (marked with blue arrows) passing by the LCA (outlined in red) at a FP prediction (ImageCAS, id: 974.img). Top: zoomed-in slices at the predicted point (red dot); bottom: adjacent slices.

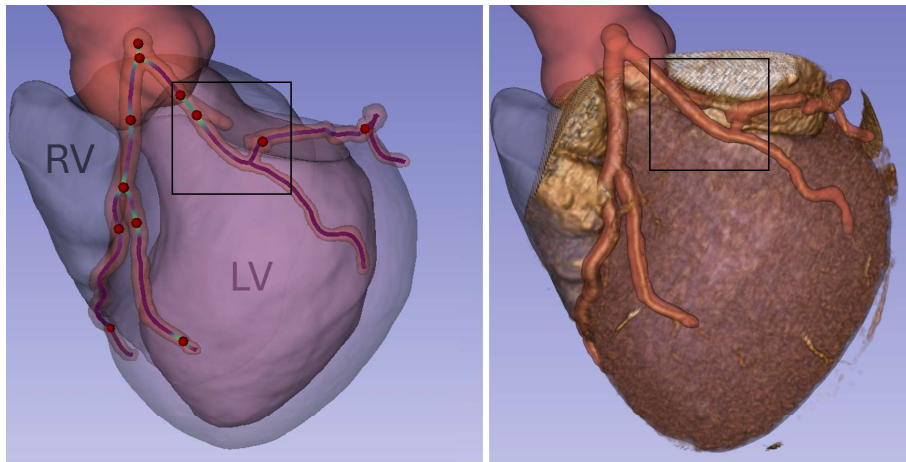


Figure C.6: Heatmap and bifurcation point prediction on an incorrect LCA annotated image (ImageCAS, id: 985.img). The volume rendering of the heart is visualized (right) to assess the annotation.